

Petro45X Transformer Oil

[Product Overview]

Petro45X transformer oil is high-grade transformer oil prepared by the deeply refined naphthenic base oil of low viscosity, low pour point and low sulfur content plus excellent antioxidant additives, and is characterized by fast heat transfer, favorable oxidization stability and excellent electric property. The product has stable and reliable properties and is produced under strict quality control procedures, and the product resources are fixed.

[Quality Level]

Petro45X transformer oil meets the requirement of I-30°C specification (special purpose) in GB2536-2011 and IEC60296-2012 standards.

[Scope of Application]

The transformer oil is used for the power distribution transformer, power generation transformer, power transformer and reactor whose LCSET is -30°C, as well as oil-filled electrical equipment with similar requirements

[Property Features]

Relatively low viscosity and viscosity index provide effective cooling and heat transfer;

Excellent thermal stability and oxidization stability prevent the oil ageing, extend the equipment service life and reduce the maintenance costs to the greatest extent;

Excellent electric insulating property, high breakdown voltage, and small DDF may effectively prevent the electric discharge and power loss in the high-voltage electric field, and meet the toughest insulating property requirements;

Excellent low-temperature property, no pour point depressant, pour point may be below -40°C, and low start temperature;

Relatively high content of cycloalkanes and appropriate content of aromatic hydrocarbons guarantee the dissolution of acidic oil sludge formed during the long-term operation or overheating failure of equipment so as not to destroy the solid insulating materials or affect the heat transfer;

Environmental-friendly and no polychlorinated biphenyl.

Petro45X transformer oil does not contain DBDS, and has passed the test by the following tough sulfur corrosion method:

----ASTMD1275 Method B

----IEC62535

----DIN51353

The additives of Petro45X transformer oil have been listed according to the requirements of

IEC60296-2012 Standard.

[Storage and Protection]

The transformer oil is easily polluted by moisture, solid particles and fiber, so it is very important to keep the transformer oil dry and clean. The storage container must be dedicated. It is prohibited to be exposed to the engine oil, hydraulic oil, gear oil and other lubricating oil. No acetylene gas is allowed in the storage container and its surroundings, and the transformer oil should be placed indoors or in the controllable climatic environment as far as possible.

[Production Place]

Taicang, Jiangsu, China

[Production Plant]

East China Lubricating Oil Plant of PetroChina

[Typical Properties]

Typical Properties of Petro45X Transformer Oil

PROPERTY	UNIT	TEST METHOD	SPECIFICATION LIMITS (IEC60296-2012)		TYPICAL DATA
			MIN	MAX	
1 –Function					
Viscosity at 40°C	mm ² /s	ISO 3104		12	9.3
Viscosity at -30°C	mm ² /s	ISO 3104		1800	1000
Pour point	°C	ISO 3016		-40	-45
Water content	mg/kg	IEC60814		30	25
Breakdown voltage					
-- Before treatment	kV	IEC60156	30		40-60
-- After treatment	kV	IEC60156	70		>70
Density at 20°C	kg/m ³	ISO 12185		895	873
DDF at 90°C		IEC60247		0.005	<0.001
2 –Refining/stability					
Appearance		IEC60296	Clear & transparent, no sediment and no suspended matter		Visual inspection
Acidity	mgKOH/g	IEC 62021-1		0.01	<0.01
Interfacial tension	mN/m	ASTMD971	40		45
Total sulfur content	%	ASTMD5453		0.15	0.01
Corrosive sulfur		DIN 51353	Non-corrosive		Non-corrosive
Potential corrosive sulfur		IEC62535	Non-corrosive		Non-corrosive
Corrosive sulfur		ASTMD1275B	-		Non-corrosive
DBDS content	mg/kg	IEC 62697-1	Not detectable		Not detectable
Inhibitors	%	IEC60666	0.08	0.40	0.38
Metal passivator additives	mg/kg	IEC60666	Not detectable		Not detectable
2-furfural content and its related compound	mg/kg	IEC61198		0.05	<0.05
3 –Performance					
Oxidation stability at 120°C,500 h		IEC61125C			
- Total acidity	mgKOH/g			0.3	0.08
- sludge	%			0.05	<0.01
- DDF at 90°C		IEC60247		0.050	0.034
4 –Health, safety and environment (HSE)					
Flash point (PM)	°C	ISO 2719	135		145
PCA content	%	IP346		3	<3
PCB content	mg/kg	IEC 61619	Not detectable		Not detectable